

Bowen Groff

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EDUCATION

University of Central Florida

Bachelor of Science in Computer Engineering

Orlando, FL

Aug 2024 – May 2028

EXPERIENCE

AMD Undergrad Research Fellow

UCF Department of Electrical and Computer Engineering

September 2026 – Present

Orlando, FL

- Incoming in UCF's iCAT lab to optimize GPU kernels and GPU-CPU systems for agentic AI and HPC workloads.

GPU Software Engineer Intern

AMD

January 2026 – May 2026

Orlando, FL

- Ported 10+ existing HIP-based GPU kernels into a testbench suite within the ROCm stack for the AMD Instinct series.
- Implemented and benchmarked async load/store, atomic, launch latency, and throughput-unrolling kernel tests, using C++20 compile-time optimization techniques to improve kernel performance by up to 73% in targeted cases.
- Implemented concurrent kernel execution across streams, increasing performance by n-fold, depending on stream count.
- Maintained HPC server infrastructure, including hardware replacement and validation of NVIDIA H100 GPUs.

Design Director

Knight Hacks

February 2026 – Present

Orlando, FL

- Leading a team of 18 designers across UI/UX, illustration, and graphic design, delegating work and providing direction.
- Developed the visual identity for KHIX, including the brand system, logo, hero artwork, and 15+ merchandise designs, driving a 3x increase in first-week applicants compared to the previous year.
- Delivered dozens of illustrations and graphics across social media, general body meetings, promotional campaigns, and event materials, reaching a community of 5,000+ Discord members and 3,000+ Instagram followers.

Software Engineer Intern

L3Harris Technologies

May 2024 – November 2025

Palm Bay, FL

- Developed a platform Linux kernel driver in C, enabling bidirectional Ethernet and Fibre Channel message transport between an FPGA and userspace over UDP, handling over 810 concurrent message buffers sorted through 8 tables.
- Mapped reserved memory blocks in custom device tree with C++ application, enabling low-latency message handling.
- Received interrupts from an FPGA and wrote custom ioctl macros to post interrupt details directly to userspace.

Software Engineer Intern

L3Harris Technologies

May 2023 – August 2023

Palm Bay, FL

- Designed and trained three predictive neural network models (including multi-layer perceptrons) using PyTorch and TensorFlow to analyze word-frequency features from a dataset of <1,000 samples, achieving ~68% prediction accuracy.
- Awarded the L3Harris RISE Award after presenting to key stakeholders for enabling F-35 ICP failure mode diagnosis ~99.5% faster, resulting in an estimated \$4M in operational savings and thousands of engineering hours saved overall.

PROJECTS

Morphogen | CUDA, C++20, CMake

July 2026 - Present

- Developing a CUDA/C++ simulation engine for GPU-accelerated mathematical pattern generation, currently supporting Mandelbrot fractal generation.
- Implemented a CUDA kernel mapping individual pixels to GPU threads to parallelize fractal computation.

Forge | Node.js, PostgreSQL, Discord API, tRPC, Zod, Cron

September 2025 - Present

- Designed and deployed a Node.js judging backend structure using tRPC and Zod for end-to-end API validation, dynamically assigning judges to hacking rooms during a live ~1,000-participant event with zero runtime failures.
- Implemented an automated cron-based role synchronization pipeline integrating PostgreSQL and Discord API, achieving ~99% reduction in manual overhead for alumni role management and ensuring consistent permissions enforcement.

SightSpeech | Python, EasyOCR, Gemini API, Flask, MediaPipe

September 2025

- Achieved 1st place out of 113 projects and 452 competitors for MLH's Best Use of Gemini API at Shell Hacks 2025.
- Built a real-time assistive OCR pipeline using EasyOCR for bounding-box word detection and Gemini API for post-OCR contextual correction and grouping, achieving ~97% word accuracy across handwritten text and mathematical expressions.
- Coordinated a split-camera architecture with MediaPipe gesture recognition in the JS frontend and EasyOCR in the Python/Flask backend, bridged via REST API to map 4 gestures to real-time text navigation and text-to-speech output.

TECHNICAL SKILLS

Languages: C, C++ (C++20), Java, Python, JavaScript, TypeScript, RTL, ARMv7 Assembly

Systems & Tools: Atlassian Suite, GitHub, Linux, Git, Docker, Jenkins, ROCm, HIP, Figma, Adobe Suite

Frameworks: React, React Native, Next.js, Node.js, Flask, Spring Boot, PyTorch, TensorFlow, Pandas, Keras